

ARCS

On the eve of the 20th century, the discovery of radiation emerging from within atoms challenged the long-held belief that atoms—which had been named for their seeming indivisibility—were the fundamental building blocks of matter. This led to the discovery of atomic nuclei and marked the beginning of the exploration of the subatomic world. At this scale, quantum physics is required to explain how natural phenomena occur. The field of nuclear physics involves the study of nuclei (the dense, positively charged objects found at the center of atoms), their constituents, and associated phenomena such as radioactivity, fission, and fusion.